AWS Certified Data Engineer Associate: Architecture Scenarios

A Visual Study Guide for Scenarios

1

Unofficial Study Guide based on 'sthithapragna' Exam Prep

NotebookLM

Serverless Redshift Access

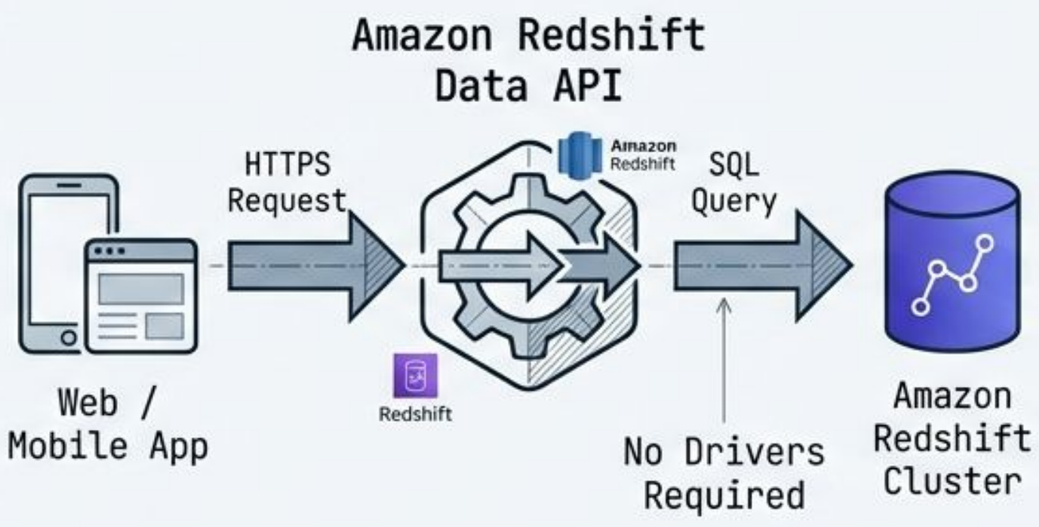
WIDTH: 300 UNIT

The Scenario

A web-based trading app needs real-time SQL queries against Amazon Redshift.

The team wants to avoid managing database connections (JDBC/ODBC) and infrastructure complexity.

The Architecture



WIDTH: 500 UNIT

The Solution

Redshift Data API

- Provides a simple HTTPS endpoint for SQL execution.
- Eliminates need for persistent connections or driver management.
- Fully managed, serverless, and ideal for web/mobile apps.

PROJECT: SERVERLESS DATA ACCESS	
DRAWING TITLE: REDSHIFT DATA API FLOW	REVISION: 1.0

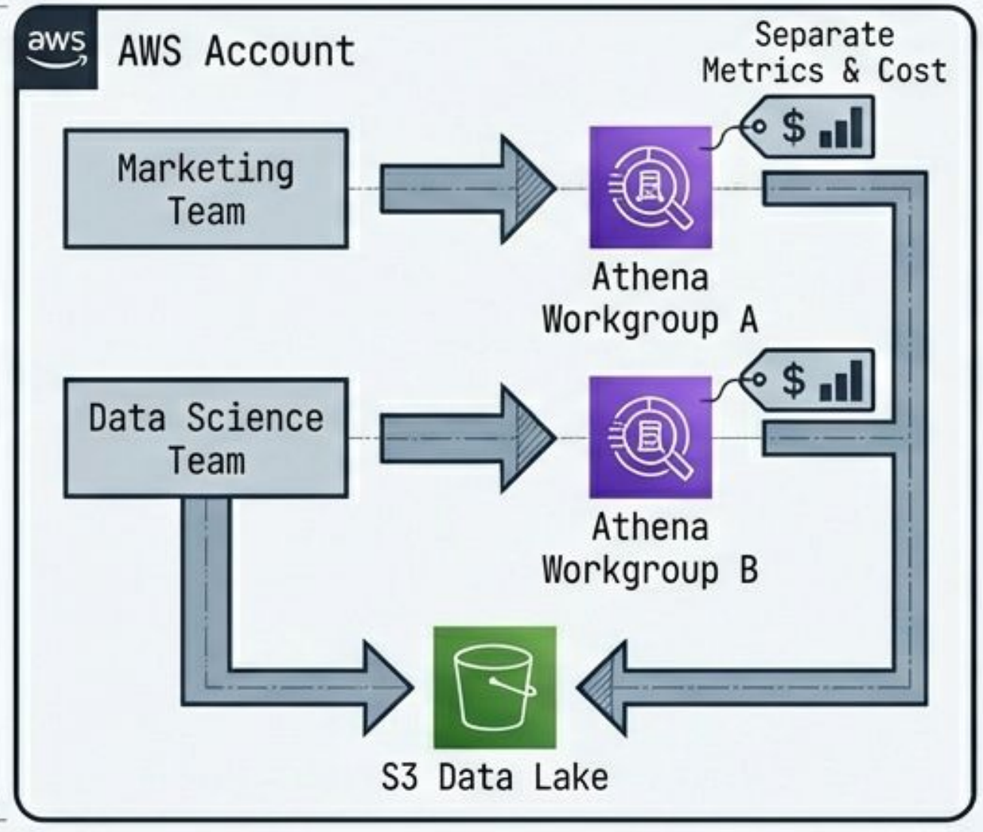
NotebookLM

Athena Multi-Tenant Governance

The Scenario

Multiple teams query data in the same AWS account. The engineer must isolate query history, permissions, and billing/cost tracking for each team.

The Architecture



The Solution

Athena Workgroups

- Logically separates users, teams, or applications.
- Each workgroup has its own query history and output location.
- Enables fine-grained cost tracking via tags and IAM policies.

PROJECT: ATHENA MULTI-TENANCY

DRAWING TITLE: WORKGROUP ISOLATION FLOW

REVISION: 1.0

NotebookLM

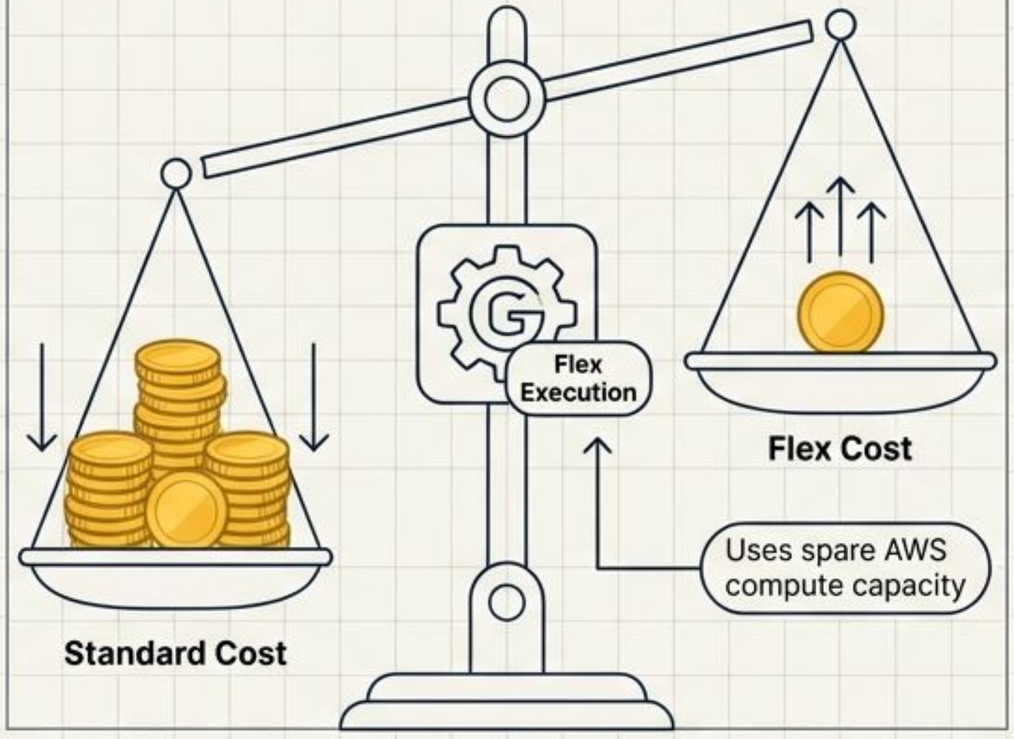
Optimizing Daily ETL

The Scenario

A daily Glue workflow runs multiple jobs.

Completion time is not critical (non-urgent), but the goal is to minimize recurring costs.

The Architecture



The Solution

Glue Flex Execution

- Offers significantly **lower costs** (up to **34% less**) for non-urgent workloads.
- Utilizes spare AWS capacity (similar to Spot instances).
- Trade-off: Start times may vary, but ideal for daily jobs with no strict SLA.

PROJECT: COST-OPTIMIZING DAILY ETL

DRAWING TITLE: FLEX EXECUTION FLOW

REVISION: 1.0

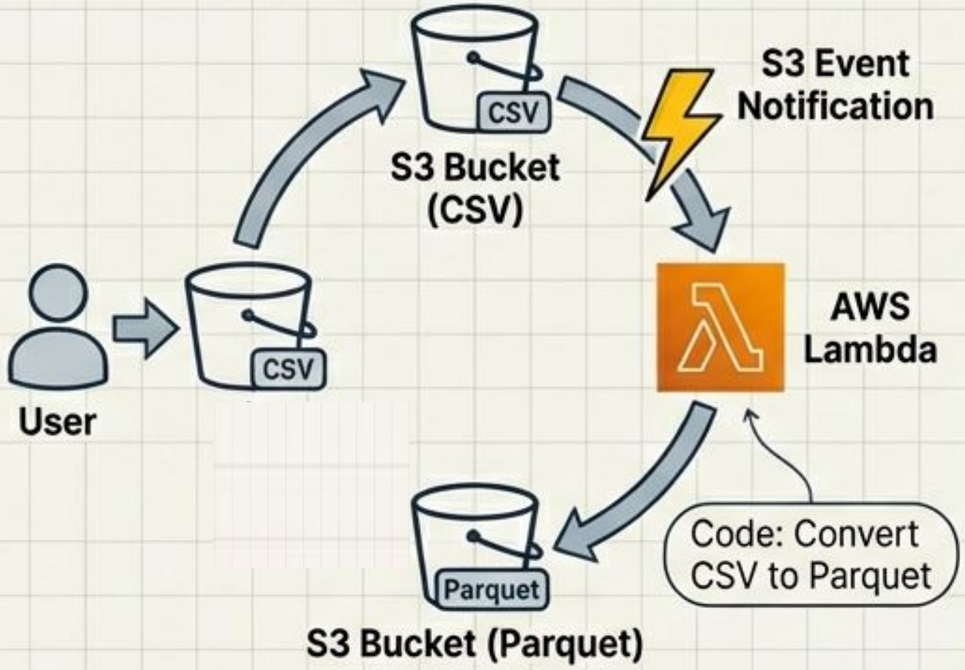
NotebookLM

Automating Data Conversion

The Scenario

Users upload CSV files to S3. The system must automatically trigger a conversion to Parquet format with zero manual intervention or server management.

The Architecture



The Solution

S3 Event Notifications + Lambda

- **Event-Driven**: Automatically triggers code when files land (filtered by suffix).
- **Serverless**: No polling or server management required.
- **Scalable**: Lambda scales automatically with upload volume.

PROJECT: COST-OPTIMIZING DAILY ETL

DRAWING TITLE: FLEX EXECUTION FLOW

REVISION: 1.0

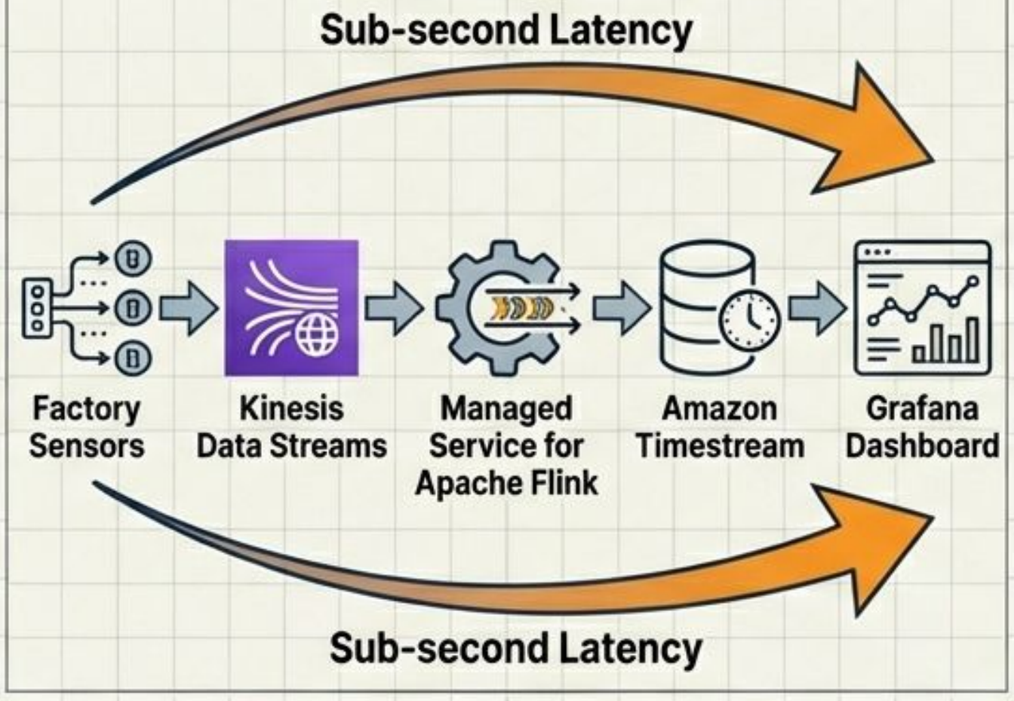
NotebookLM

Low-Latency Factory Monitoring

The Scenario

A factory collects continuous sensor data via Kinesis. Needs to calculate and visualize metrics on a display with minimal latency (real-time).

The Architecture



The Solution

Amazon Managed Service for Apache Flink

- Processes streaming data in real-time with low latency.
- Stateful processing capabilities (windows, aggregations).
- Integrates natively with Kinesis and Timestream for time-series visualization.

Grafana enables near real time visualization

PROJECT: LOW-LATENCY FACTORY MONITORING
DRAWING TITLE: REAL-TIME DATA PIPELINE
REVISION: 1.0

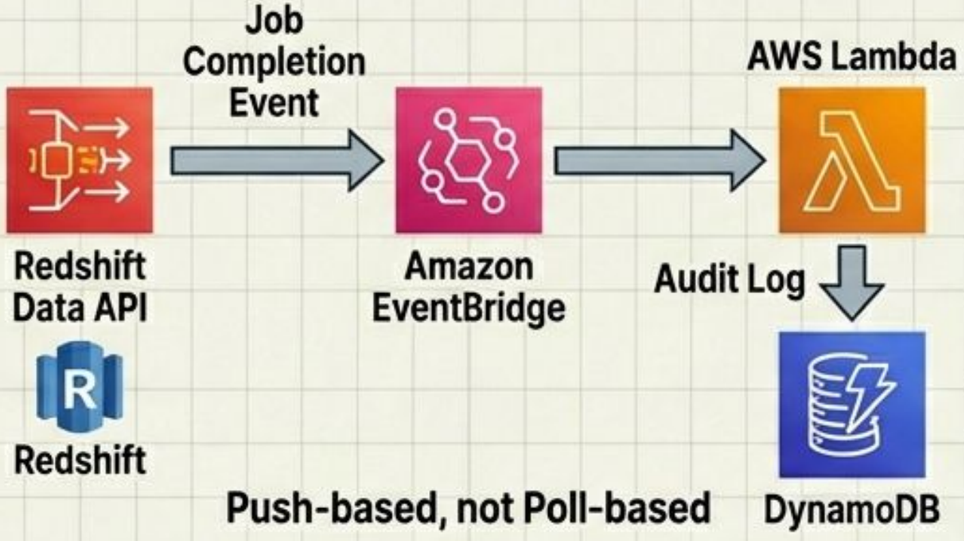
NotebookLM

Auditing Redshift Loads

The Scenario

Track the success/failure status of Redshift load jobs. Status must be automatically logged to DynamoDB for auditing without polling.

The Architecture



The Solution

EventBridge & Redshift Data API

- **Redshift Data API**: Publishes events upon query completion.
- **EventBridge**: Acts as the orchestration layer to route specific events.
- **Lambda**: Parses the event and writes metadata to DynamoDB.

PROJECT: REDSHIFT LOAD AUDITING

DRAWING TITLE: EVENT-DRIVEN AUDIT FLOW

REVISION: 1.0

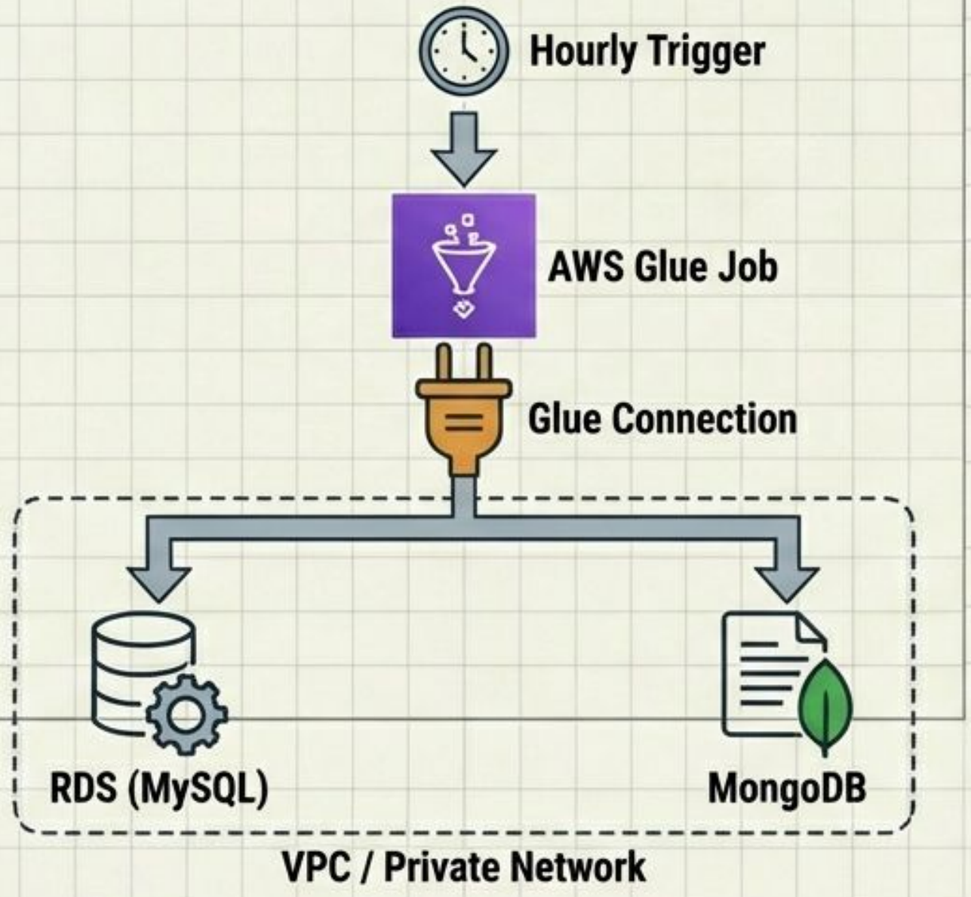
NotebookLM

Secure ETL Connectivity

The Scenario

A Glue ETL pipeline needs to extract data from RDS and MongoDB (inside a VPC), transform it, and load it into Redshift. It requires secure connections.

The Architecture



The Solution

AWS Glue Connections

- Stores connection properties (VPC, security groups, credentials) securely.
- Allows seamless access to multiple data stores (JDBC/MongoDB).
- Managed triggers handle the hourly scheduling automation.

PROJECT: SECURE ETL CONNECTIVITY

DRAWING TITLE: GLUE VPC INTEGRATION

REVISION: 1.0

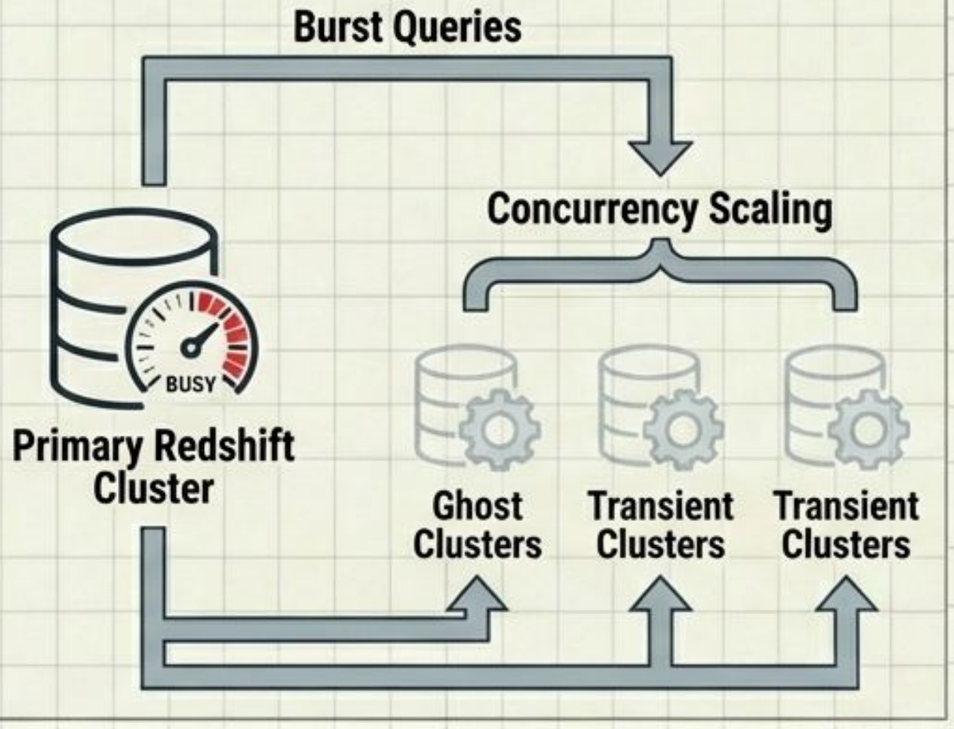
NotebookLM

Handling Query Spikes

The Scenario

A Redshift cluster (RA3 nodes) experiences performance degradation during occasional heavy query bursts due to limited concurrency.

The Architecture



The Solution

Redshift Concurrency Scaling

- Automatically adds transient capacity to handle bursts in query load.
- Enabled at the Workload Management (WLM) queue level.
- Ensures consistent performance during spikes without resizing the main cluster.

PROJECT: REDSHIFT CONCURRENCY SCALING

DRAWING TITLE: CONCURRENCY SCALING ARCHITECTURE

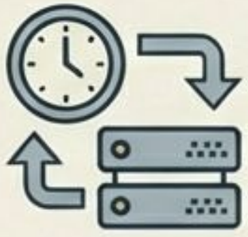
REVISION: 1.0

NotebookLM

Orchestrating Long Queries

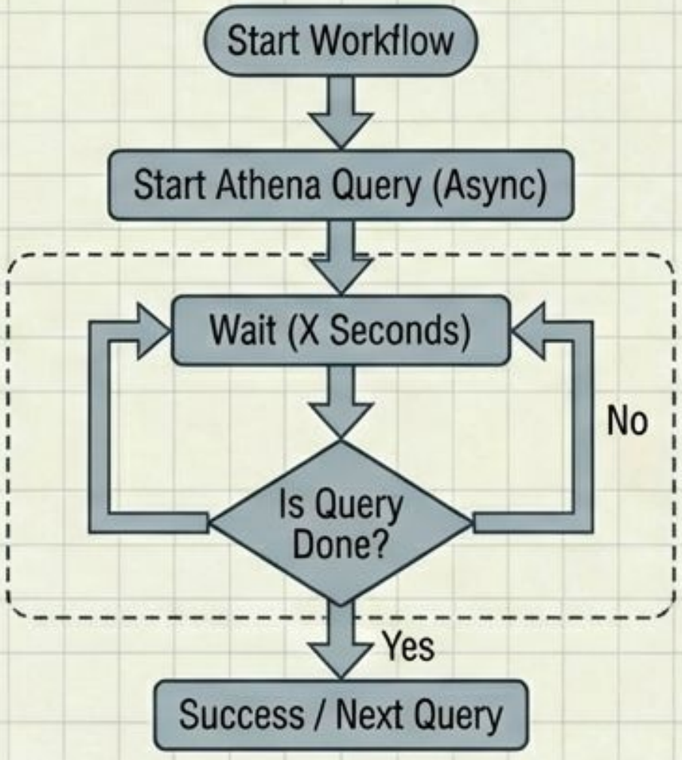
The Scenario

A daily workflow runs multiple Athena queries sequentially. Some queries take >15 minutes. The solution must be serverless and automated.



The Architecture

The Polling Pattern



The Solution

AWS Step Functions

- Manages state and workflow for long-running processes.
- Overcomes Lambda's 15-minute timeout limit.
- Triggers Athena queries and implements a 'wait & check' pattern.

PROJECT: ORCHESTRATING LONG QUERIES

DRAWING TITLE: AWS STEP FUNCTIONS POLLING PATTERN

REVISION: 1.0

NotebookLM

Scenario: Transactional Data Lakes

The Challenge

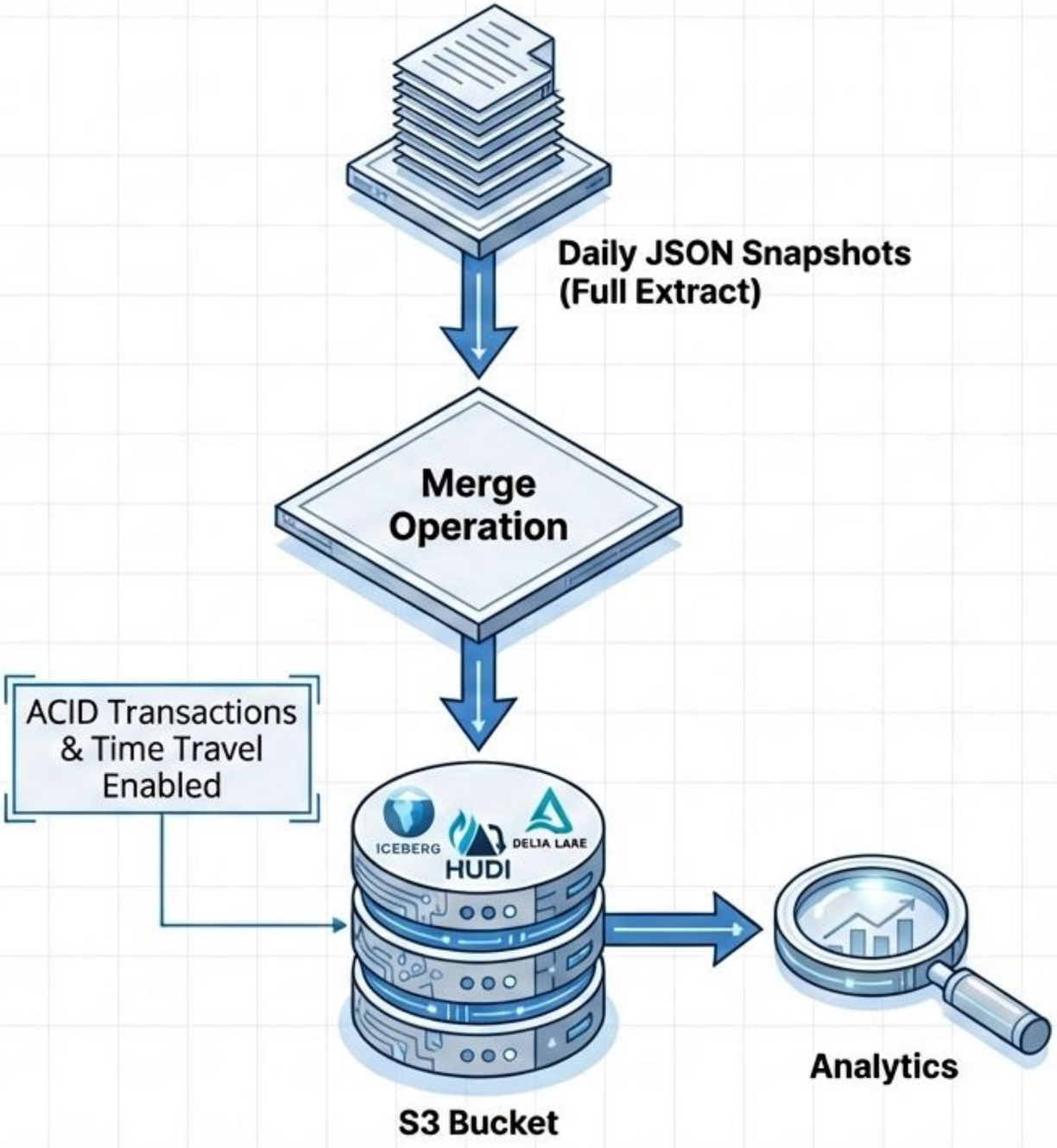
A data lake receives daily JSON snapshots with full extracts. Identifying and applying only changed records (CDC) efficiently is difficult without complex processing.

The Solution

Modern Open Data Lake Formats
(Iceberg, Hudi, Delta Lake)

The Exam Key

Native support for ACID transactions, Merge-on-Read, and Time Travel. Eliminates complex Lambda logic for diff-ing records.



NotebookLM

Scenario : Redshift Performance Tuning

The Challenge

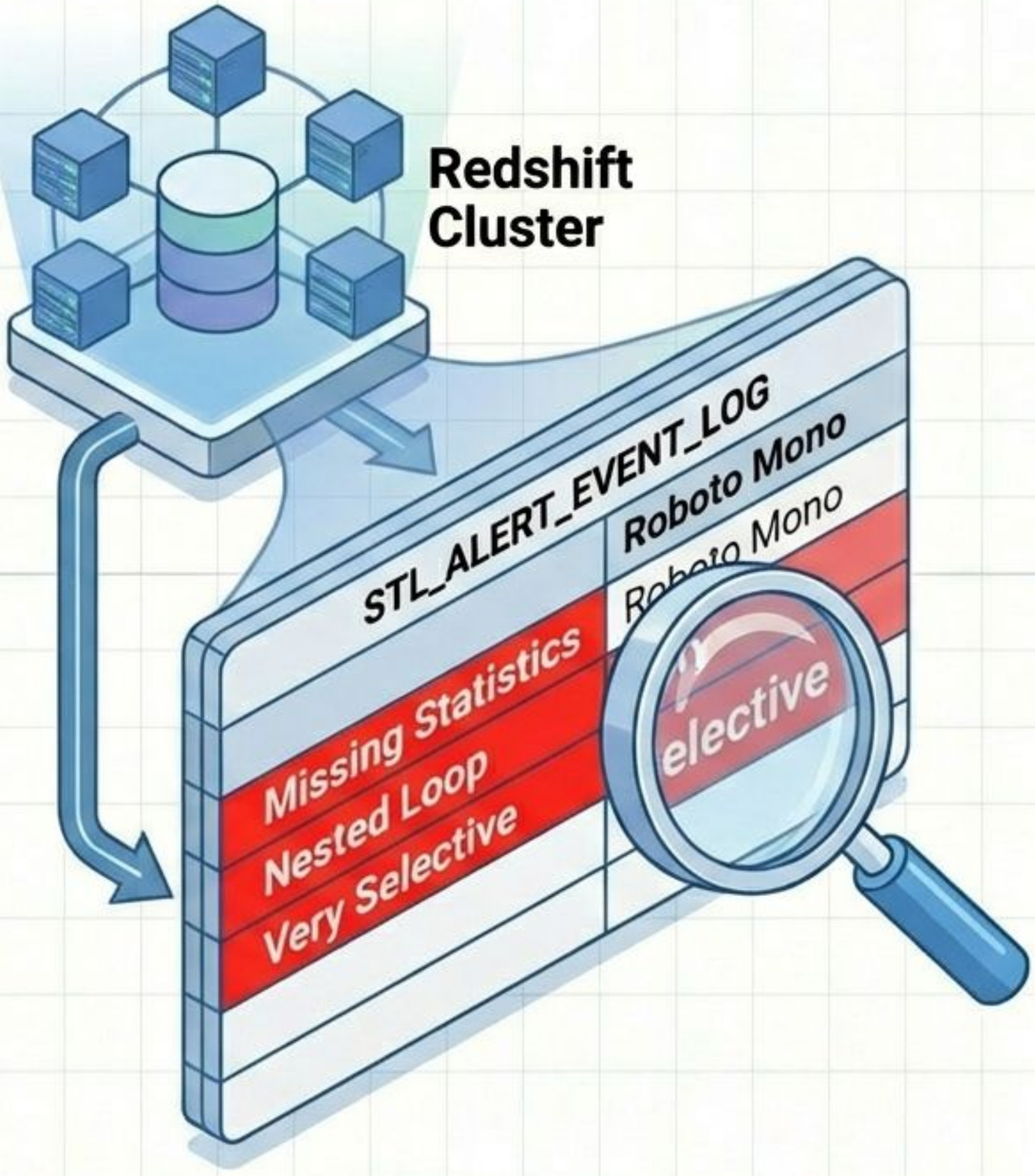
Detect specific query anomalies such as missing statistics, skewed joins, or nested loops for performance tuning.

The Solution

STL_ALERT_EVENT_LOG

The Exam Key

This specific system table records optimizer alerts. It is the primary source for diagnosing execution anomalies versus general query history.



NotebookLM

Scenario : Optimization for Athena

The Challenge

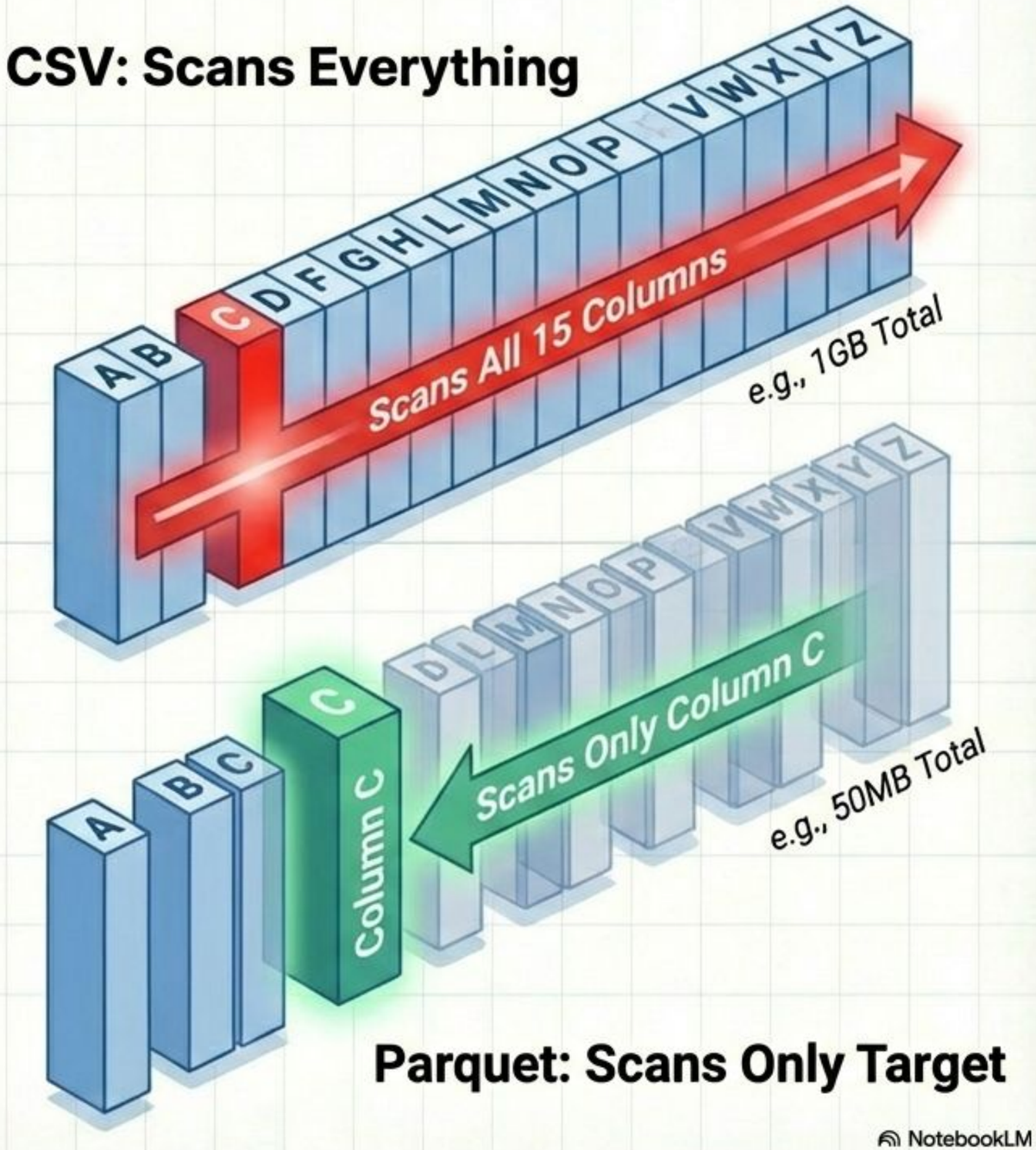
Minimizing data scan costs when analysts only query 1 or 2 columns from large 15-column CSV files.

The Solution

Apache Parquet (Columnar Format)

The Exam Key

Parquet stores data by column. Athena scans *only* the requested columns (e.g., 50MB) instead of the whole file (e.g., 1GB), reducing cost significantly.



NotebookLM

Scenario : Data Governance

The Challenge

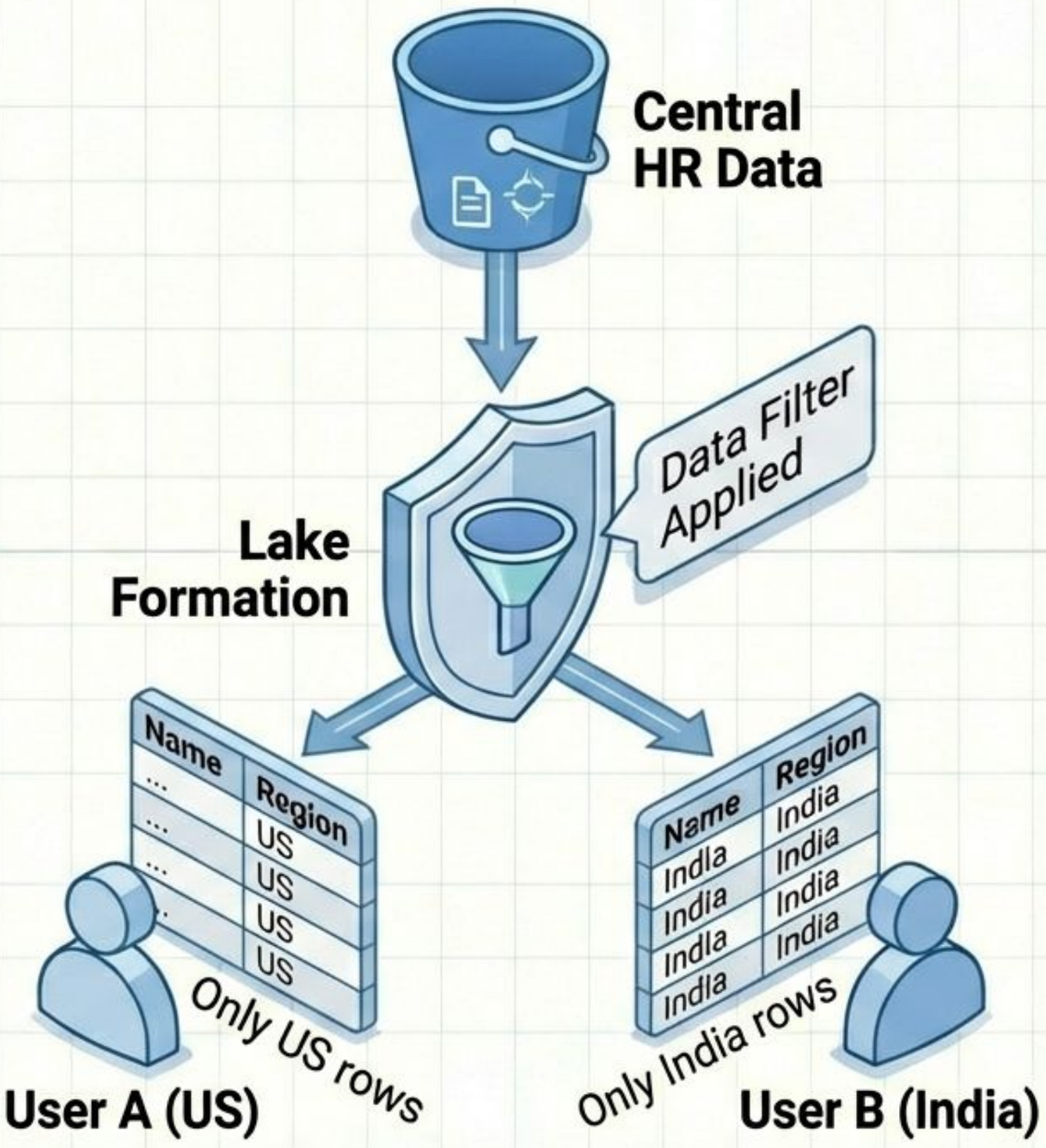
Central S3 data lake. Regional HR teams (US, India) must only see employee data from their specific region. Minimize storage duplication.

The Solution

Register the S3 path as a data location in AWS Lake Formation and enable fine-grained access control with region-based data filters for each HR IAM role.

The Exam Key

Use **Data Filters** for Row-Level Security. Apply filters based on column values (e.g., `Region='US'`) dynamically at query time.



NotebookLM

Scenario : Orchestration Troubleshooting

The Challenge

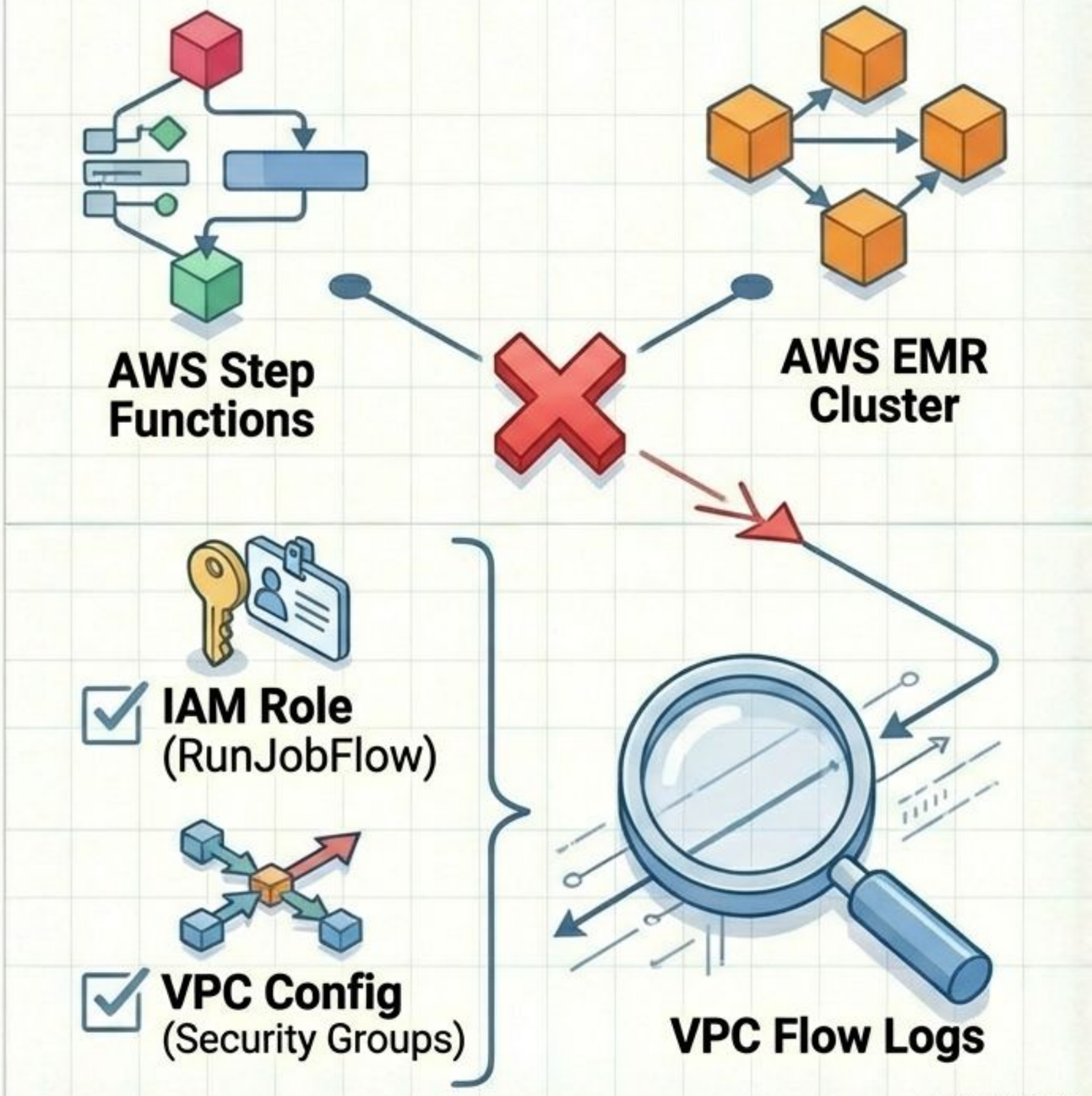
A Step Functions workflow fails to start an EMR job. Need to identify the root cause of the connection failure.

The Solution

Check IAM Policies & VPC Network

The Exam Key

1. **IAM:** Ensure Role has `elasticmapreduce:RunJobFlow`.
2. **Network:** Check Security Groups and VPC Flow Logs for blocked traffic.



NotebookLM

Scenario: Athena Partitioning

The Challenge

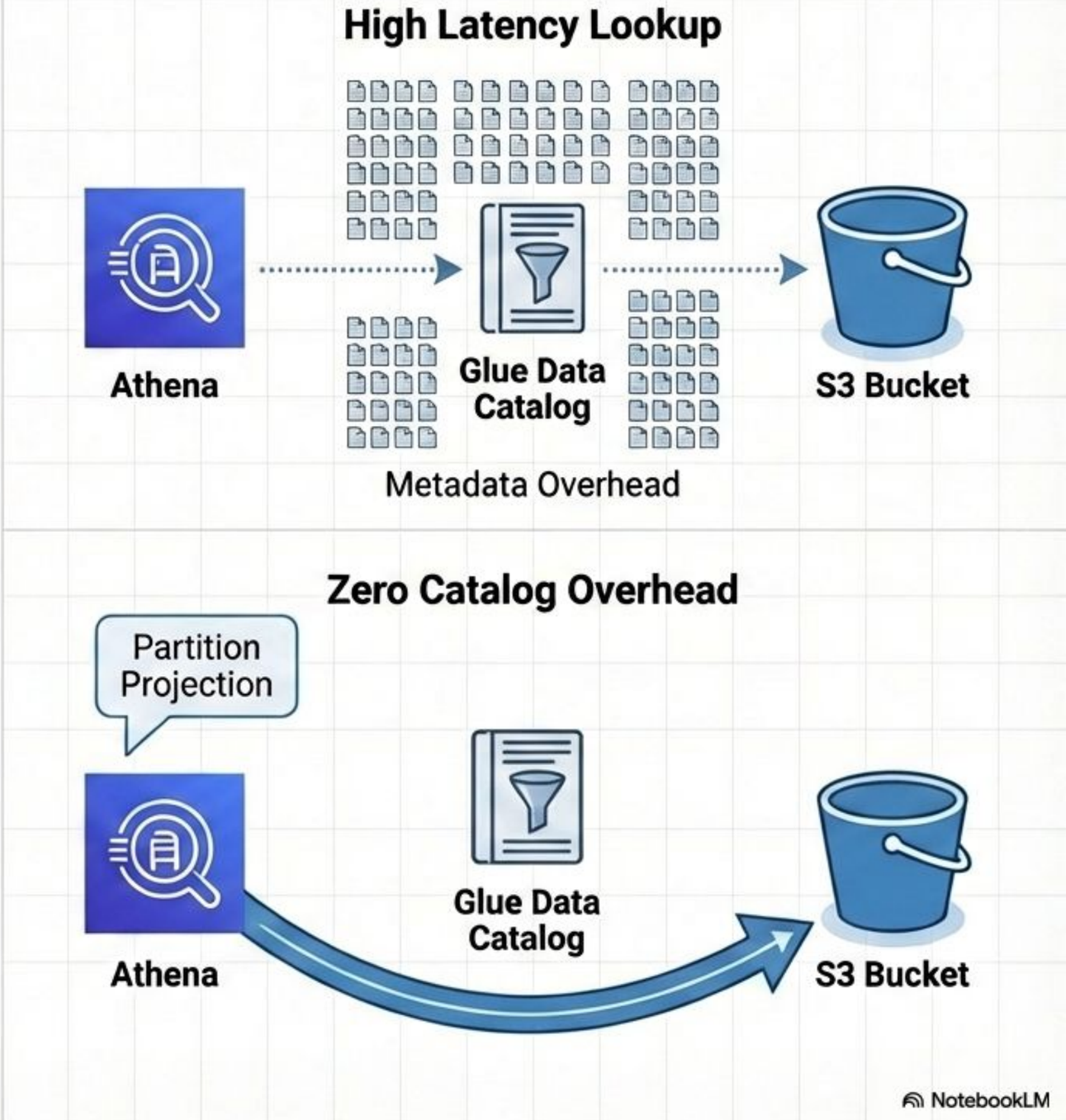
Athena queries are slow due to high planning time (latency) when retrieving metadata for thousands of S3 partitions from the catalog.

The Solution

Partition Indexes & **Partition Projection**

The Exam Key

Partition Indexes speed up pruning. Partition Projection bypasses the Glue Catalog entirely by calculating partition locations algorithmically.



NotebookLM

Scenario: Managed Streaming Analytics

The Challenge

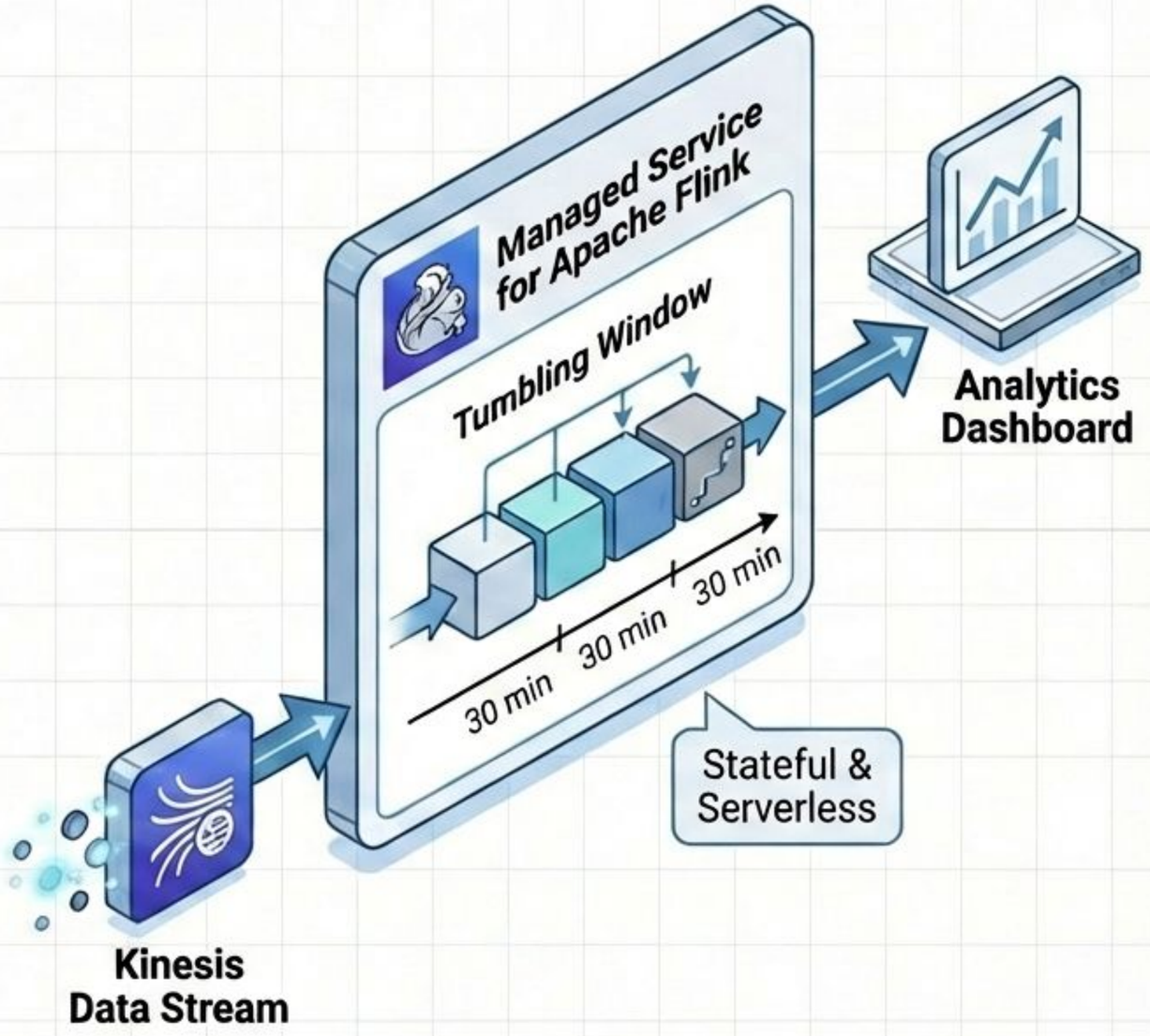
Perform real-time time-based aggregations (e.g., 30-minute windows) on streaming data with high fault tolerance and no server management.

The Solution

Amazon Managed Service for Apache Flink

The Exam Key

Native support for stateful processing, tumbling/sliding windows, and checkpointing. Much more capable for windowing than Lambda.



NotebookLM

Scenario: EBS Volume Upgrades

The Challenge

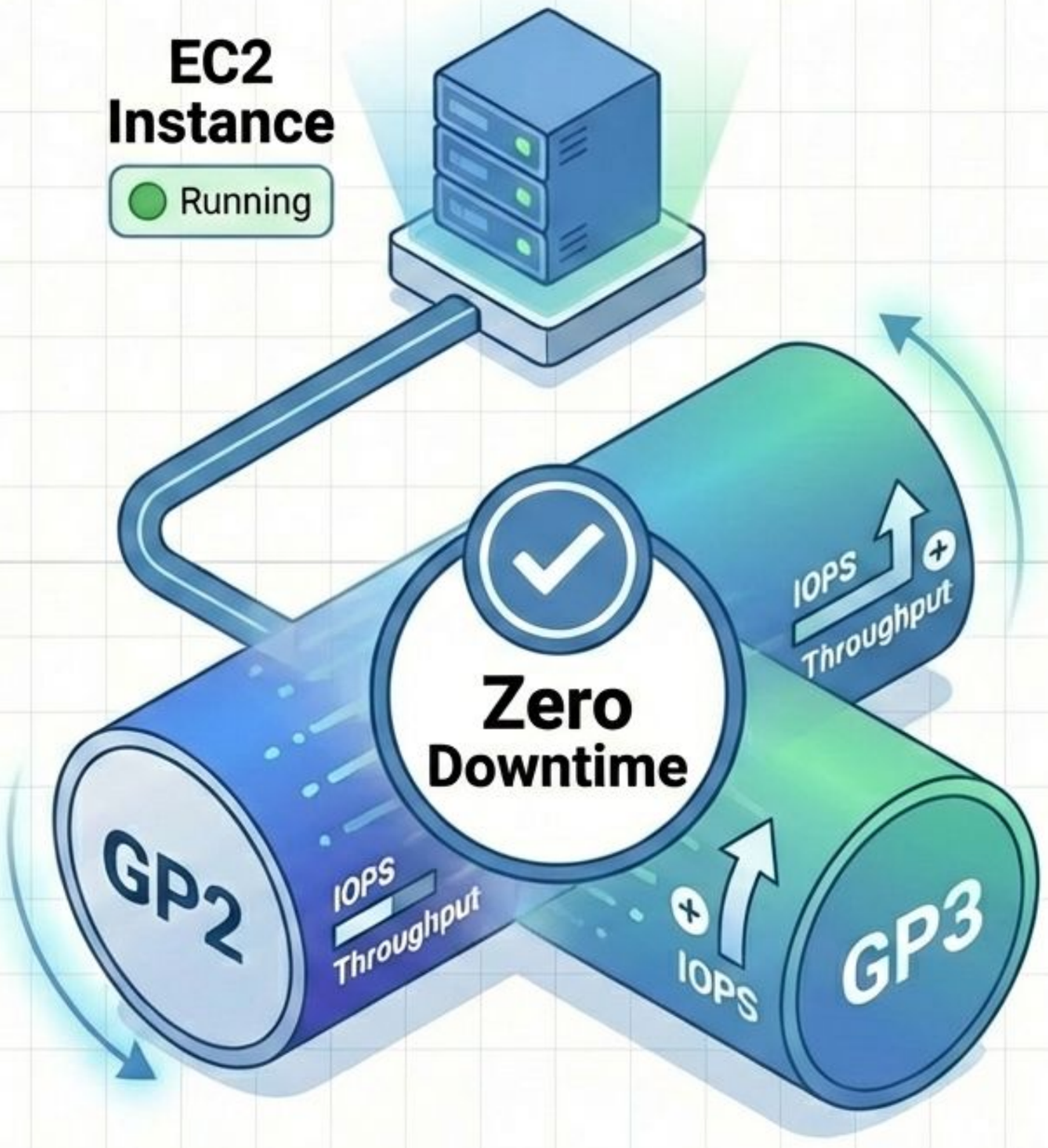
Upgrade EC2 block storage from GP2 to GP3 for cost/performance optimization without stopping the instance or detaching volumes.

The Solution

Elastic Volumes (Modify Volume)

The Exam Key

EBS modification is an online operation. Zero downtime. Zero data loss. Changes size, type, and IOPS on the fly.



NotebookLM

Scenario: SQL Server to S3 Parquet

The Challenge

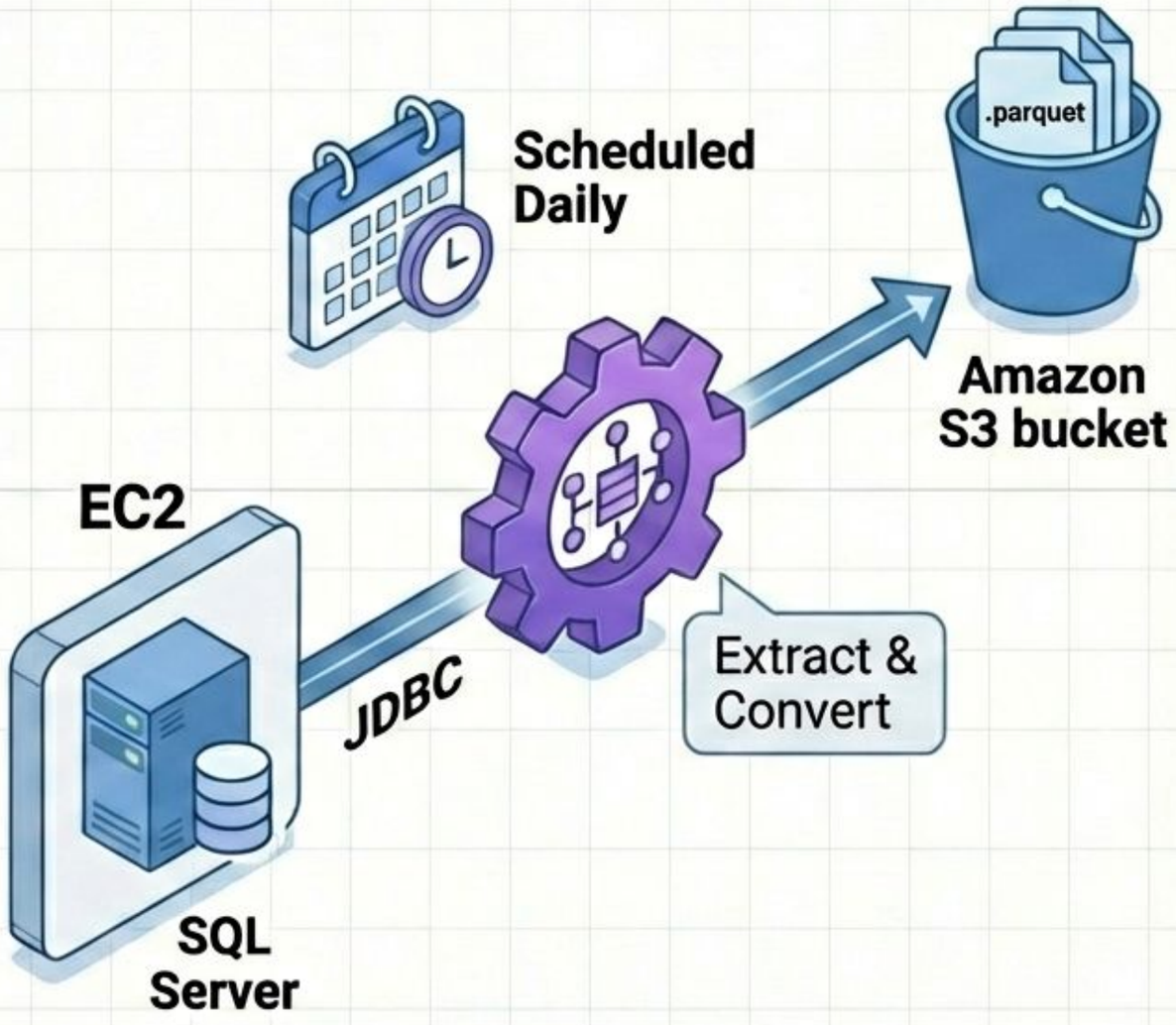
Automate the daily export of joined datasets from a legacy SQL Server on EC2 to S3 in Parquet format.

The Solution

AWS Glue ETL Job

The Exam Key

Glue connects via JDBC. It handles complex SQL queries, infers schema, and natively converts output to Parquet without custom scripting.



NotebookLM

Scenario : Spark on Athena

The Challenge

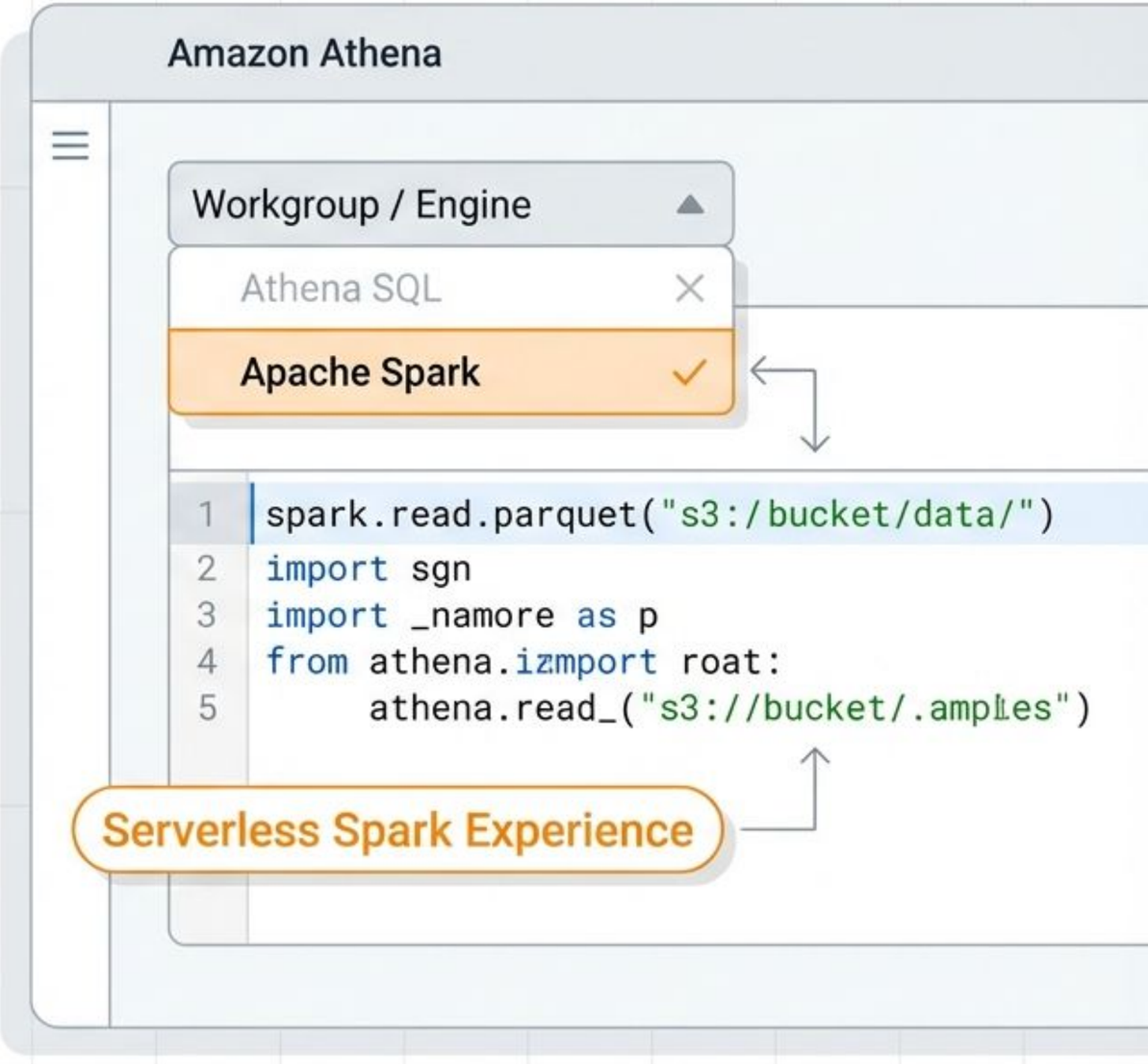
Team wants to run Apache Spark analytics (Python/Scala) but wants to use the familiar Athena interface without managing EMR clusters.

The Solution

Athena for Apache Spark

The Exam Key

Run interactive Spark applications directly in Athena Workgroups. Serverless, instant startup (<1s), no infrastructure to manage.



NotebookLM